Real-Time Marathi Sign Language Recognizer Using Machine Learning

Rohini Rai
dept. of Computer Engineering
(Bachelor's of Engineering)
Terna Engineering College:
Navi Mumbai, Maharashtra, India
rairohini2001@gmail.com

Arya Sachin Patil
dept. of Computer Engineering
(Bachelor's of Engineering)
Terna Engineering College:
Navi Mumbai, Maharashtra, India
aryasachinpatil286@gmail.com

Avantika Gharat
dept. of Computer Engineering
(Bachelor's of Engineering)
Terna Engineering College:
Navi Mumbai, Maharashtra, India
avantikagharat777@gmail.com

Mahii Variar
dept. of Computer Engineering
(Bachelor's of Engineering)
Terna Engineering College:
Navi Mumbai, Maharashtra, India
mahiivariar26@gmail.com

Prof. Pramila Mate
dept. of Computer Engineering
(Assistant Professor)
Terna Engineering College:
Navi Mumbai, Maharashtra, India
pmate1513@gmail.com

Abstract—**Our project focuses on creating a special system that can understand and interpret Marathi Sign Language movements in real-time. It then changes these movements into written words, which are spoken out loud using a natural-sounding voice. This technology helps people who use sign language to communicate with those who do not use the same, making communication easier and more inclusive for everyone.**

Index Terms—Image Capturing, Hand detecting and tracking, Segmentation, Pre-processing, Feature Extraction, Training and Testing, Recognition, Final Output, Marathi alphabets Dataset, Hand Gestures, Camera.

I. Introduction

Let us say you would like to speak with a deaf individual. This may already seem like a challenging task, especially if you are not familiar with the sign language. Millions of deaf people struggle with this issue since it prevents them from having conversations with hearing people. The ability to communicate is crucial to human existence. It is difficult to detect the expressions without communication which seems the main task.

Our project facilitates communication by recognizing various hand gestures symbols, and actions using a webcam and converting them into meaningful messages and further in speech format. The sign language recognizer primarily recognizes Marathi Sign Language, which includes the Marathi alphabet and translates additional words into text or voice. The system manages two-hand gestures, allowing the user to operate it openly as well as enhancing accuracy using the trained and tested dataset. Marathi Sign Language Recognizer detects signs automatically from hand expressions and recognizes them accordingly. This sign language Recognition involves converting the signs such as hand gestures and interpreting them to assist close the communication gap with hearing-impaired persons. This system will assess the signals, actions, and motions and categorize them into messages using a machine-learning technique.

A technique that recognizes Marathi Sign Language and converts it to text will be more helpful for these individuals so they can communicate more effectively with outsiders. In MSLRs, words and sentences are formed by static hand morphologies called postures, hand movements called
gestures, which correlate to words and sentences. The principal features of this system consist of its ability to identify two-handed gestures and convert them into either text or audio. This Marathi Sign Language Recognizer employs both hand gestures in place of existing ISL, which focuses solely on one hand movement, to improve the number of user interfaces. As a result, it is necessary to examine the specifics of sign descriptions to accurately comprehend them and make judgments about the key details.

II. Literature Survey

In paper [1] authors Amrutha K, Prabhu P. studied the pre-processing, segmentation, feature extraction, classification, and recognition stages systematically. After a comparison of various ML-based methods, they selected the Convex Hull method for feature selection. However, the accuracy yielded through this experiment was just 65%. Less accuracy was obtained when the gestures were performed continuously at a fast pace and the small size of the dataset used. The paper [2] mainly emphasizes the use of the Marathi language for dataset creation. The use of deep learning was made to translate the Marathi sign languages into

Marathi alphabets (Swar and Vyanjan). Also, the system provided around 95% accuracy using right-hand and left-hand gestures simultaneously.

The paper [3] module is used to map the spoken alphabet to text with high accuracy. This paper also offers dynamic gesture detection without loss of accuracy. Although there are several advantages this paper offers, improvement is needed when there is a lot of cutter and poor lighting in the environment. In the paper [4] the authors used an HMM chain for each gesture and a k-NN model to classify each hand pose. The result was recognition of hand pose in about 0.2s and that for gesture was 0.0037s in real time which was very efficient. Although the paper should include a large and accurate sign language dataset.

One of the best ways to communicate freely is without pretending and without changing ourselves as per the system's environment. The system should be able to identify our hand gestures naturally. This same concept was implemented successfully in paper [5] where bare hands and facial features were decoded smoothly. The concept of considering facial features, and emotions while detecting a sign was stated in the paper [6] by authors Razieh Rastgoo, Kourosh Kiani, and Sergio Escalera. The system stated in the paper was never implemented thus the success rate of this system is not known.

III. Overview of Marathi Sign Language

The Marathi Sign Language consists of Swar and Vyanjan as shown in fig(3.1). So we created our dataset considering the Marathi Sign Swar and Vyanjan alphabets as mentioned below:

अ आ इ ई उ ऊ ए ऐ ओ औ

क ख ग घ ङ च छ ज झ ञ

ट ठ ड ढ ण त थ द ध न

प फ ब भ म य र ल व

श ष स ह ळ

Fig 3.1 Marathi Alphabets

We further created our dataset and listed it in the image format as shown in below Figures (3.2), (3.3) and (3.4) it contains a total of 43 letters and hand gestures. With this data, the words are recognized and generate the output.



Fig. 3.2. Marathi language Dataset (I)

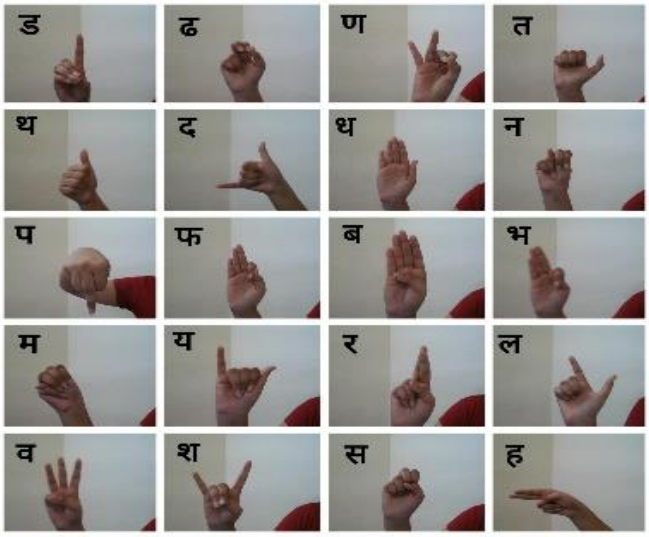


Fig. 3.3 Marathi language Dataset (II)



Fig. 3.4 Marathi language Dataset (III)

The 21 landmark points is considered as shown below in below figure (3.5), would be strategically placed on the hand to represent key features of the signs. These landmark points could include positions on the fingers, palm, and other relevant regions of the hand.

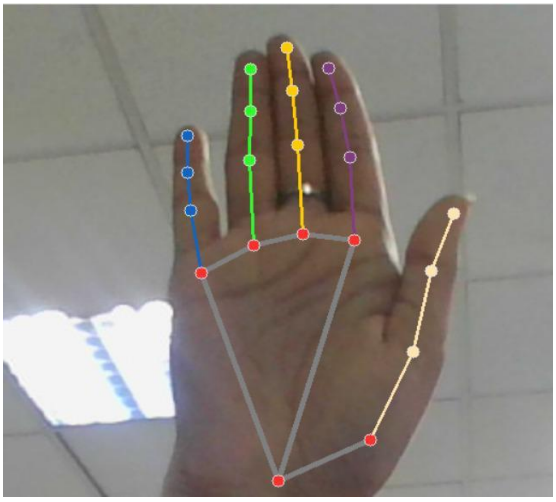


Fig. 3.5. Landmark points detection on hand in real time

IV. RELATED WORK

The project focuses on the creation of a Marathi Sign Language Recognizer using Machine Learning in Real Time. The proposed System focuses on identifying the hand gestures through the Web Camera, further, the gestures are converted into meaningful text or speech format. The system is trained and tested with various algorithms to provide the best accu-racy and performance measures. Its hand landmarks points represent the key features of the signs including the positions of fingers, palm and so on. The feature selection of the hand gestures the pre-processing, and extraction is done. After the completion of processing, the sign picture for that text appears on the screen. With the use of an existing database, patterns are recognized and matched during the translation of sign language to text and vice versa. Furthermore, when using a web camera for a Marathi sign language recognizer, an image of a hand gesture is obtained by the device and analyzed to determine the proper text for the associated input gesture picture. The difficult aspect of the suggested system is identifying the input hand gesture image. From the input that the system is trained on, the proposed method will determine the appropriate output. The dataset is evaluated and trained using a variety of algorithms to improve accuracy, and the final output is recognized with a precise frequency. Thus, our system consists of a user interface (UI) that displays text output and a textbox that generates voice with just a click of a button.

V. PROPOSED SYSTEM

i.Capture image from camera:
The user gives the required input through the web camera and it must be in proper format eliminating the unwanted noise from the background so that the proper image will be easily detected.

ii. Pre-processing:
In order to facilitate frequent communication between users, the system focuses on primary hand gestures and movements.
Pre-processing is carried out when a text or picture is entered. It loads the input into the system. After obtaining this input, the system processed it for feature extraction.

iii.Feature Extraction:

Variables: The dataset includes coordinates and temporal data of key points extracted from the MediaPipe framework. These points represent various body parts and movements involved in signing gestures.

Data Types and Format: The data is primarily composed of numerical values representing the spatial coordinates of key points extracted from the videos. The dataset is stored in a structured format compatible with MediaPipe's output specifications.

Data Size and Characteristics: The dataset consists of 'N' recordings, each containing 'M' frames, where each frame captures the spatial coordinates of 'K' key points. The dataset has been reviewed for any inconsistencies or missing values, ensuring its quality for subsequent analysis.

iv. Creating Hand Gestures for Dataset Designing
The dataset for the Marathi sign language recognizer project was generated by us using the MediaPipe framework. Due to the unavailability of Marathi sign language datasets online, we took the initiative to create our dataset. The collected data is primarily, obtained through video recordings of individuals performing various Marathi sign language gestures. These recordings were then processed using MediaPipe to extract key points and form the dataset.

v.Loading images for Training
The image is trained using the cleaned data. The dataset is trained to obtain performance measures such as accuracy to generate the correct output.

vi.Training Dataset
Training data is essential for teaching our system to identify patterns and enhance its hand gesture performance to produce text or speech. Almost 80% of the data is used to train the model.

vii.Testing Dataset
While in testing the dataset only 20% of data is used. Data is tested to obtain the validation to evaluate the performance.

viii.Output:
The output of this model or a system is a label or a text that is predicted and shown at the user interface by matching the

hand gesture shown by user via webcam with the images from the created dataset. It also converts the text into speech format with the help of the button which is displayed at the user interface and it totally depends on the user if they want to excess the text-to-speech functionality or not.

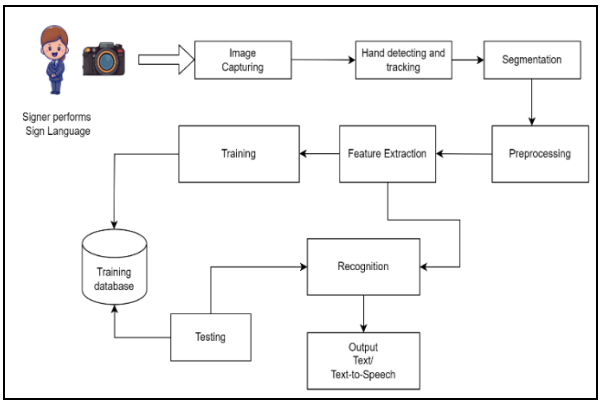


Fig. 3.5. Methodology

VI. RESULT AND ANALYSIS

Result and analysis depicts the exact working or the functionality and the terms which are considered during the execution of the application or system after the completion. The screenshots given below depicts the idea about how Marathi Sign Language Recognizer works at the user interface :



Fig. 3.6. User Interface (I)

The above figure (3.6) shows the User Interface that pops up after running the application.

It is very simple interface because as it a software based application therefore the focus is not on designing the user interface, our focus was on achieving the accuracy.

If the application is not able to detect the hand gestures or the landmarks then it will give an error message saying that “Hand landmarks not detected”.



Fig. 3.7. User Interface (II)

In the above figure (3.7), if I am showing any gesture which does not exists in the dataset then it will give a message saying that “Hand gesture not predicted”.



Fig. 3.8. User Interface (III)

In the above figure (3.8) , if the system correctly predicts the hand gesture which is available in the dataset then it will show a label or a text corresponding to that image from the dataset.



Fig. 3.9. User Interface (IV)

Similarly, it can also predict the words or phrases which are available in the dataset.

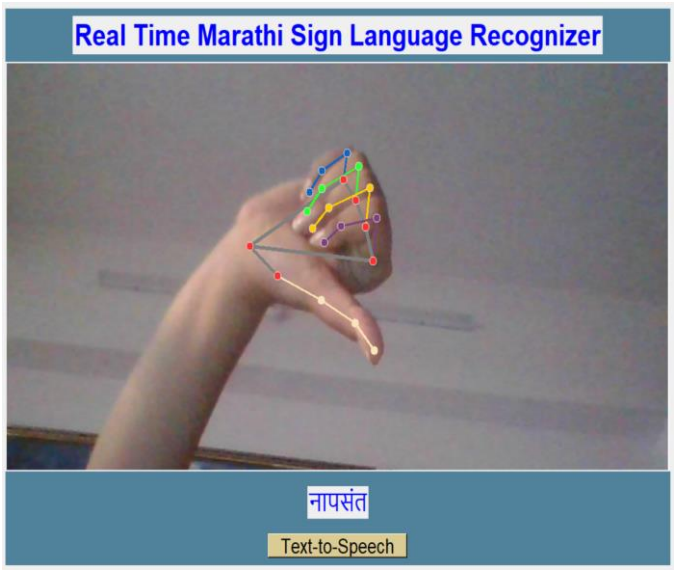


Fig. 3.10. User Interface (V)



Fig. 3.11. User Interface (VI)

The user can also convert the text into speech format by clicking on the “Text-to-Speech” button.

Analysis :

While we were developing this project then we faced a lot of challenges and among all those challenges the biggest challenge was how to achieve an accuracy so that the system should correctly identify or predict the text from the dataset.
In order to achieve an accuracy we thought that aur dataset is very small or it lack the variations with respect to colour, hand size, orientation etc. So by keeping all these things in mind we extended our dataset and yes the accuracy increased but not to the extent which we were in need of ,so our next option was to train our model by using various algorithms and do a comparative analysis of those algorithms and select a best algorithm which fits to the model.

Algorithm	Accuracy score	Precision score	Error score
CNN (Convolutional Neural Network)	98.38%	0.97	1.16%
SVM (Support Vector Machine)	96.30%	0.95	3.50%
Random Forest Algorithm	94.40%	0.93	4.95%
KNN (K-Nearest Neighbors)	91.56%	0.90	8.45%

Fig. 3.12. Result Analysis table

The above figure (3.12), shows the comparison of the algorithms which best fits the model with respect to their Accuracy score, Precision score and Error Score.
From this we concluded that CNN (Convolutional Neural Network) gives the best accuracy which is "98.38%".
Also CNN was a suitable algorithm for our dataset because our dataset contains an images.

VII. CONCLUSION AND FUTURE WORK

The Marathi Sign Language Recognizer is like a smart tool that helps deaf people who use Marathi Sign Language communicate better. It can turn their signs into spoken words, making it easier for them to talk to people who don't understand sign language. It's like a bridge between two different ways of communicating. Additionally, it can also read written Marathi text out loud, which can be helpful for people who have trouble reading or seeing. So, it's not only for the deaf but also for others who may need assistance with reading. This technology is a big step towards making communication easier and more inclusive for everyone.

In conclusion, we accomplished an impressive accuracy for our Marathi Sign Language recognition system. Furthermore, we have shown that our system can process signs in real-time, calculating milliseconds per frame. The system's features, which are and respectively, show how well it can identify Marathi Sign Language gestures while reducing false positives. The Marathi Deaf community's accessibility and communi-cation could be greatly enhanced by these findings. There are obstacles to overcome, though, such as variances in sign language and possible cultural disparities. Ongoing initiatives should concentrate on dataset expansion, user education, and cross-cultural adaptation to optimize the system's impact. Through the integration of technology and a dedication to inclusivity, the Marathi Sign Language will be enhanced.

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